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H SECTION
OS-MINIPROJECT

1.Multi-container supervision

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ./roo
tfs-alpha "/bin/sleep 60"
started container 'alpha' pid=35247
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta ./root
fs-beta "/bin/sleep 60"
started container 'beta' pid=35307
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start gamma ./roo
tfs-alpha "/bin/echo hello_logging"
started container 'gamma' pid=35347
```

2.Metadata tracking

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE       STARTED             EXIT
gamma           35347   exited      2026-04-14 20:21:28 0
beta            35307   running     2026-04-14 20:20:44 0
alpha           35247   exited      2026-04-14 20:20:07 0
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```

3.Bounded-buffer logging

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine logs gamma
child_fn: mount proc (warning): No such file or directory
hello_logging
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```

4.CLI and IPC

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE       STARTED             EXIT
gamma           35347   exited      2026-04-14 20:21:28 0
beta            35307   exited      2026-04-14 20:20:44 0
alpha           35247   exited      2026-04-14 20:20:07 0
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine logs beta
child_fn: mount proc (warning): No such file or directory
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```

5.Soft-minit warning

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start soft ./root
fs-alpha "./memeory_hog" --soft-mib 50
started container 'soft' pid=35830
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail -20
[40514.241021] audit: type=1400 audit(1776198265.427:1243): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40519.800609] audit: type=1400 audit(1776198270.987:1244): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40530.215609] audit: type=1400 audit(1776198281.403:1245): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40535.218653] audit: type=1400 audit(1776198286.403:1246): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40540.219637] audit: type=1400 audit(1776198291.407:1247): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40545.220464] audit: type=1400 audit(1776198296.407:1248): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
```

```
[40580.297809] audit: type=1400 audit(1776198331.483:1255): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40585.299641] audit: type=1400 audit(1776198336.483:1256): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40590.300665] audit: type=1400 audit(1776198341.487:1257): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40595.302015] audit: type=1400 audit(1776198346.487:1258): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40602.408496] audit: type=1400 audit(1776198353.591:1259): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40690.515879] audit: type=1400 audit(1776198441.698:1260): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40695.534888] audit: type=1400 audit(1776198446.718:1261): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40755.153213] [container_monitor] Registering container=soft pid=35830 soft=524
28800 hard=67108864
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```

6.Hard-limit enforcement

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start hard ./root
fs-alpha "./memory_hog" --hard-mib 100
started container 'hard' pid=36040
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail -20
[40780.948363] audit: type=1400 audit(1776198532.129:1262): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40810.274505] audit: type=1400 audit(1776198561.461:1263): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40831.544259] audit: type=1400 audit(1776198582.729:1264): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40836.546579] audit: type=1400 audit(1776198587.733:1265): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40841.547823] audit: type=1400 audit(1776198592.729:1266): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40860.516828] audit: type=1400 audit(1776198611.703:1267): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40881.492818] audit: type=1400 audit(1776198632.680:1268): apparmor="DENIED" op
```

```
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40922.843110] [container_monitor] Registering container=hard pid=36040 soft=419
43040 hard=104857600
[40927.047257] audit: type=1400 audit(1776198678.240:1277): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40932.047335] audit: type=1400 audit(1776198683.240:1278): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40937.048907] audit: type=1400 audit(1776198688.240:1279): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[40942.049875] audit: type=1400 audit(1776198693.245:1280): apparmor="DENIED" op
eration="open" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=3
059 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE       STARTED             EXIT
hard            36040   exited      2026-04-14 20:31:14 127
soft            35830   exited      2026-04-14 20:28:26 127
gamma           35347   exited      2026-04-14 20:21:28 0
beta            35307   exited      2026-04-14 20:20:44 0
alpha           35247   exited      2026-04-14 20:20:07 0
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```

7.Scheduling experiment

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start fast ./root
fs-alpha "./cpu_hog" --nice -5
started container 'fast' pid=36256
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine start slow ./root
fs-alpha "./cpu_hog" --nice 10
started container 'slow' pid=36339
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ ps -o pid,ni,comm
    PID  NI COMMAND
  35015   0 bash
  36355   0 ps
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```

8.Clean teardown

```
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine stop beta
Error: container 'beta' is not running (state=exited)
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
Error: container 'alpha' is not running (state=exited)
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ sudo pkill -f engine
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$ ps aux | grep engine
spoorthi   36649  0.0  0.1   6480  2288 pts/2    R+   20:39   0:00 grep --color=
auto engine
spoorthi@spoorthi-vm:~/OS-Jackfruit/boilerplate$
```